

Formative Assignment
CSE211s Introduction to Embedded Systems
Spring 2026

Q1. Classify the following Thumb instructions into their correct categories:

Data Movement, Arithmetic, Logical, Branch, Compare/Test, Shift/Rotate.

LDR R0, [R1]
ADD R2, R3, R4
BNE LOOP
MOV R5, #10
SUBS R6, R6, #1
CMP R0, R1
ANDS R2, R2, R3
LSL R4, R4, #2
BL DELAY

Q2. Write a program that:

Given:

R0 = 0x0F0F0F0F

R1 = 0x00FF00FF

Perform:

1. AND operation → store in R2
 2. OR operation → store in R3
 3. XOR operation → store in R4
 4. Logical shift left R0 by 4 → store in R5
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Q3. Write a program to:

1. Compare two numbers stored in R0 and R1
2. If R0 > R1 store 1 in R2
3. If R0 < R1 store -1 in R2
4. If equal store 0 in R2

Addressing modes

Q1.

List four addressing modes available in the Thumb instruction set and briefly explain each.

Q2.

What is PC-relative addressing in Thumb instruction set?

Q3.

Given:

MOVS R1, #25

ADDS R2, R1, #15

(a) Identify the addressing mode used.

(b) What is the final value of R2?

Q4.

Assume:

- R3 = 0x20000000
- Memory[0x20000000] = 0x55

Instruction:

LDRB R0, [R3]

(a) What addressing mode is used?

(b) What is the value of R0 after execution?

Q5.

Assume:

- R4 = 0x20000000
- Memory[0x20000008] = 0x12345678

Instruction:

LDR R1, [R4, #8]

(a) What addressing mode is used?

(b) Calculate effective address.

(c) Value loaded into R1?

Q6.

Given:

- R5 = 0x20000000
- Memory[0x20000004] = 0xAA

Instruction:

LDRB R0, [R5, #4]!

- (a) What addressing mode is used?
- (b) What is the final value of R5?
- (c) What is R0?

Q7.

Given:

- R6 = 0x20000000
- Memory[0x20000000] = 0x11

Instruction:

LDRB R0, [R6], #4

- (a) Identify addressing mode.
- (b) Value loaded into R0?
- (c) Final value of R6?

Q8. Compare Immediate and Register Addressing**Q9. Array Access in Thumb**

Assume:

- R0 contains base address of integer array
- Each integer = 4 bytes
- R1 contains index i

Write a Thumb instruction to load array[i] into R2.

Q10. PC-relative Addressing Numerical Problem

Assume:

- PC = 0x08000000

Instruction:

LDR R0, [PC, #12]

Calculate effective address.

Q11. Why are multiple addressing modes important in embedded systems?

Model Answer:

Multiple addressing modes:

1. Reduce instruction count
2. Improve execution efficiency
3. Allow direct array and structure access
4. Enable efficient stack operations
5. Support pointer-based programming in C

In microcontrollers like STM32F401RE, efficient addressing improves:

- Code density
- Execution speed
- Power consumption

Q12. Stack Operation Addressing Mode

Explain the addressing mode used in:

PUSH {R4-R7}

POP {R4-R7}